Flow Control and Error Control

Data Communications and Networking, 3rd Edition, Behrouz A. Forouzan

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1 Flow and Error Control

Flow and error control are two important functions of the **Data Link Layer**. They ensure that data is transmitted between the sender and receiver efficiently and reliably.

1.1 Flow Control

Definition: Flow control is a set of procedures used to control the amount of data that a sender can transmit before it must wait for an acknowledgment from the receiver.
Why is Flow Control Needed? The sender and receiver may operate at different speeds. For example, a sender may transmit data much faster than the receiver can process it. If the sender continuously transmits data at a high rate, the receiver's buffer may become full and some data may be lost. Therefore, flow control prevents the sender from overwhelming the receiver.
Receiver Buffer: A receiver generally has a temporary memory area called a **buffer**. The receiver stores incoming data in the buffer before processing it. If the incoming data rate is higher than the processing rate, the buffer starts filling. When the buffer becomes nearly full, the receiver must ask the sender to reduce or temporarily stop transmission.

1.2 Error Control

Definition: Error control is a mechanism used to detect and recover from errors that occur during data transmission.
In the data link layer, the term *error control* refers primarily to methods of error detection and retransmission. Error control in the data link layer is often implemented simply: Anytime an error is detected in an exchange, specified frames are retransmitted. This process is called **automatic repeat request (ARQ)**.
Flow and Error Control Mechanisms: In subsequent sections we introduce three common flow and error control mechanisms:

1. Stop-and-Wait ARQ
2. Go-Back-N ARQ
3. Selective-Repeat ARQ

Although these are sometimes referred to as protocols, we prefer the term *mechanisms*.

2 Stop-and-Wait ARQ

Stop-and-Wait ARQ is the simplest flow and error control mechanism. It provides reliable transmission by allowing the sender to send one frame and wait for an acknowledgment before sending the next frame.

2.1 Basic Principle

In Stop-and-Wait ARQ, the sender follows these basic steps:

- The sender sends a frame to the receiver.
- The sender keeps a copy of the last transmitted frame.
- The sender waits for an acknowledgment (ACK).
- If the correct ACK is received, the sender sends the next frame.
- If the frame or ACK is lost or damaged, the sender retransmits the frame.

Thus, at any particular time, only one data frame is outstanding.

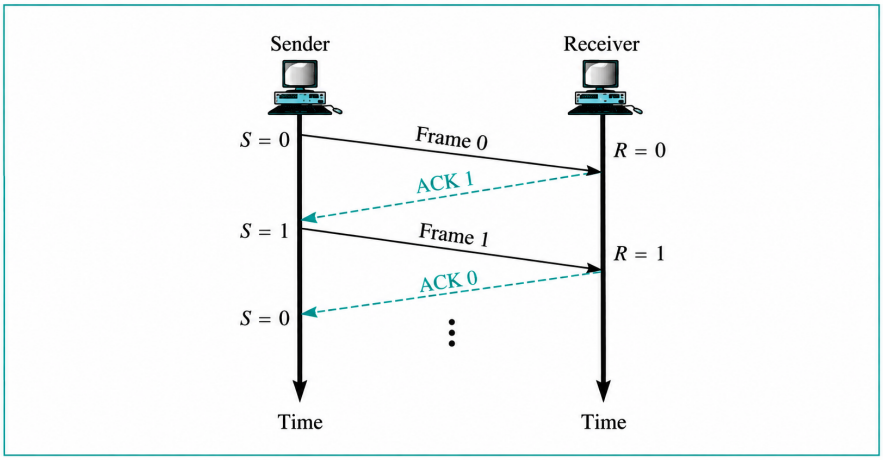
The sender keeps a copy of the last frame transmitted until it receives an acknowledgment for that frame. Keeping a copy allows the sender to retransmit a lost or damaged frame until it is received correctly. Both data frames and acknowledgment (ACK) frames are numbered alternately using **0 and 1**. A data frame numbered 0 is acknowledged using ACK 1, which indicates that frame 0 has been received and the receiver is now expecting frame 1. A data frame numbered 1 is acknowledged using ACK 0, which indicates that frame 1 has been received and the receiver is now expecting frame 0. The alternating sequence numbers help the receiver identify duplicate transmissions. If an acknowledgment is lost or delayed, the sender may retransmit the same frame. The receiver can identify the duplicate using the sequence number.

Stop-and-Wait ARQ uses two sequence numbers: **0 and 1**. The sender uses a control variable called S, which indicates the sequence number of the frame being sent. The receiver uses a control variable called R, which indicates the sequence number of the next frame expected.

2.2 Operations

Normal Operation: In normal transmission, the sender sends Frame 0 and waits to receive ACK 1. When ACK 1 is received, the sender sends Frame 1 and then waits to receive ACK 0.

Figure 11.1 *Normal operation*



The sender and receiver use the control variables S and R, respectively.

- S represents the sequence number of the frame being sent.
- R represents the sequence number of the next frame expected by the receiver.
- Initially, $S = 0$ and $R = 0$.
- After receiving Frame 0, the receiver sends ACK 1 and expects Frame 1.
- After receiving Frame 1, the receiver sends ACK 0 and expects Frame 0.

Lost or Damaged Frame: A lost or damaged frame is handled in the same way by the receiver.

If the receiver receives a damaged frame, it discards the frame and does not send an acknowledgment.

If a frame is lost during transmission, the receiver remains silent because it does not receive the frame.

The sender starts a timer when it sends a frame. If an acknowledgment is not received within the allotted time period, the sender assumes that the frame was lost or damaged and retransmits it.

Lost Acknowledgment

Figure 11.2 *Stop-and-Wait ARQ, lost frame*

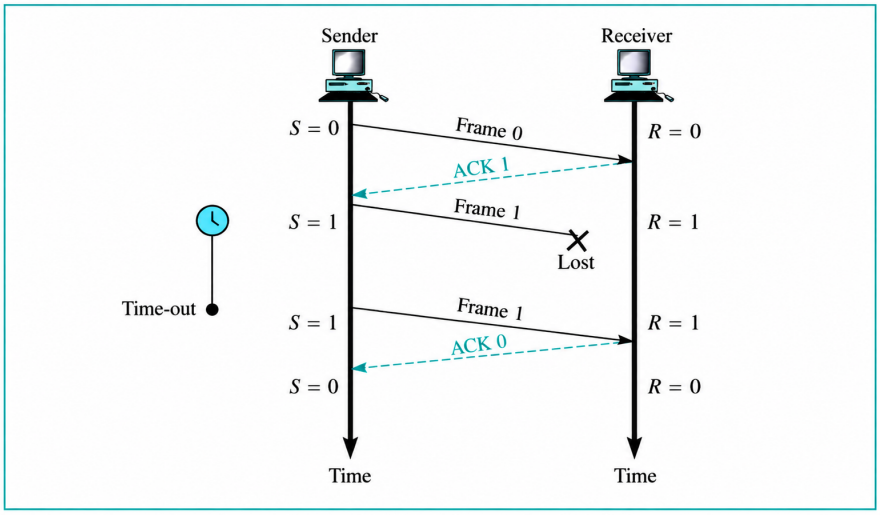
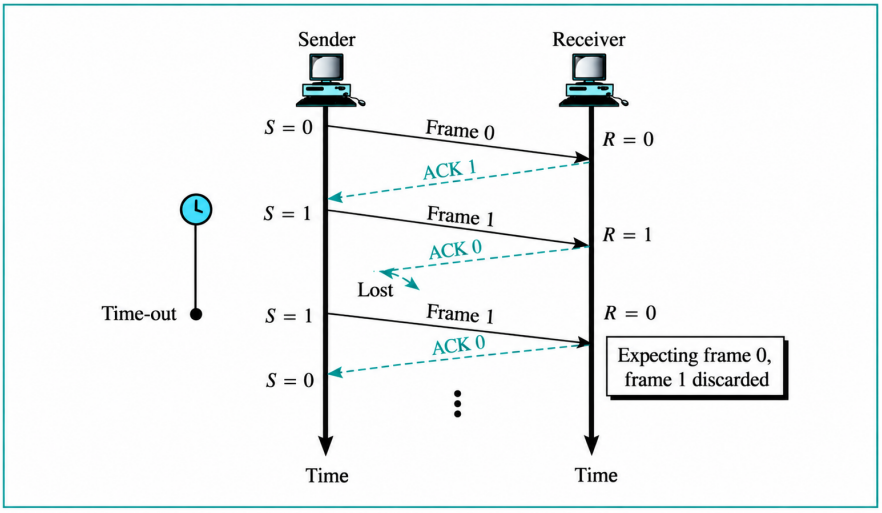


Figure 11.3 *Stop-and-Wait ARQ, lost ACK frame*



A lost or damaged acknowledgment is handled in the same way by the sender.

If the receiver receives a frame correctly, it sends an acknowledgment. However, if the acknowledgment is lost or damaged, the sender does not receive it.

The sender's timer eventually expires. The sender then assumes that the frame was lost or damaged and retransmits the frame.

The receiver may receive the retransmitted frame as a duplicate.

The receiver uses the sequence number to identify the duplicate and discards it.

If the receiver has already received the original frame, it recognizes the retransmitted frame as a duplicate.

Delayed Acknowledgment

An acknowledgment may sometimes be delayed because of processing or transmission delay.

Suppose the sender sends Frame 0 and the receiver successfully receives it. The receiver sends ACK 1, but the acknowledgment is delayed.

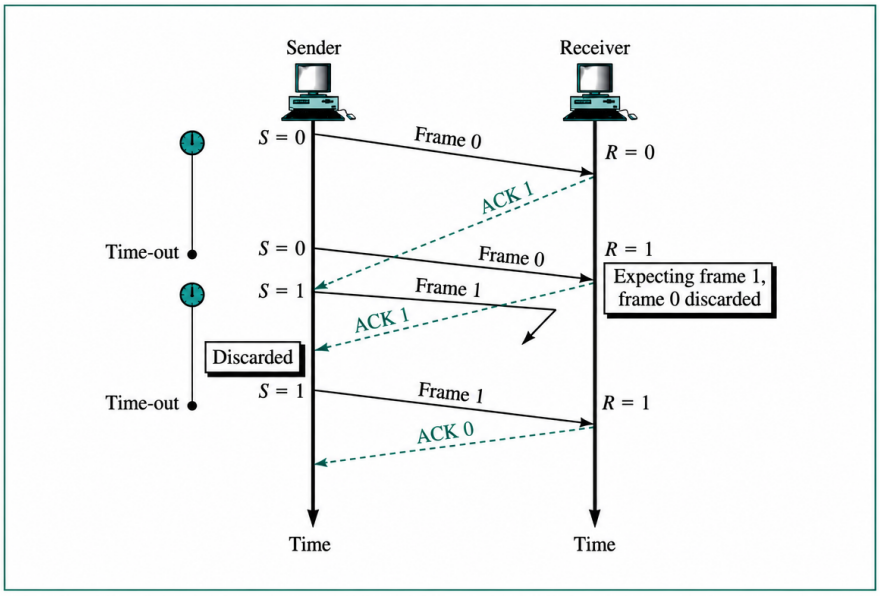
Meanwhile, the sender's timer may expire and the sender may retransmit Frame 0.

The receiver has already received Frame 0 and is expecting Frame 1. Therefore, the retransmitted Frame 0 is recognized as a duplicate and is discarded.

The receiver may nevertheless send another ACK 1.

Thus, numbered acknowledgments are necessary when an acknowledgment is delayed and the next frame is lost or delayed.

Figure 11.4 *Stop-and-Wait ARQ, delayed ACK*



3 Go-Back-N ARQ

Go-Back-N ARQ is a more efficient flow and error control mechanism than Stop-and-Wait ARQ. In Stop-and-Wait ARQ, at any point in time for a sender, there is only one frame outstanding. The sender must wait for the acknowledgment before transmitting the next frame.

To improve efficiency, it is desirable to allow more than one frame to be outstanding while waiting for acknowledgments.

Go-Back-N ARQ is one of the mechanisms that allows multiple frames to be transmitted before receiving acknowledgments.

The sender keeps a copy of the transmitted frames until the corresponding acknowledgments arrive. This procedure requires additional variables compared with Stop-and-Wait ARQ.

3.1 Sender and Receiver Sliding Window

Sequence Numbers: Frames from a sending station are numbered sequentially.Because the sequence number is included in the frame header, the number of bits used for the sequence number determines the number of available sequence numbers.

If m bits are used for the sequence number, the sequence numbers are

$$0, 1, 2, \ldots, 2^m - 1.$$

After the sequence number $2^m - 1$, the sequence number is repeated starting from 0.

Therefore, the sequence numbers are used modulo 2^m.

For example, if $m = 3$, the available sequence numbers are0,1,2,3,4,5,6,7.

After sequence number 7, the sequence starts again from 0: 0,1,2,3,4,5,6,7,0,1,2,3,.....

Sender Sliding Window

At the sender, the outstanding frames are the frames for which acknowledgments have not yet been received. To keep track of these frames, the sender uses a **sliding window**.

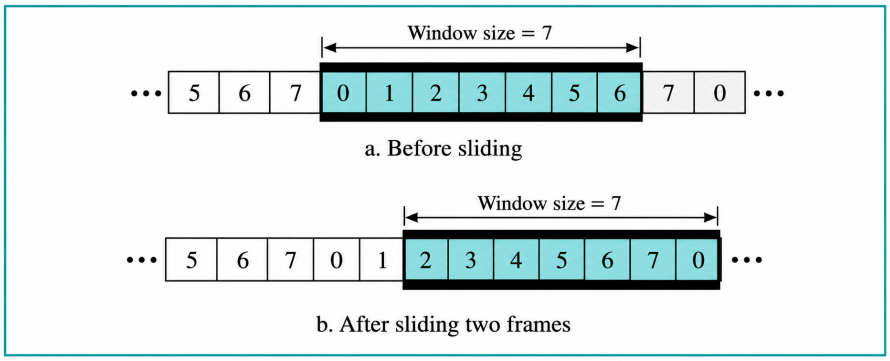
The window represents the range of sequence numbers that can currently be used by the sender.

The frames inside the sender window can be transmitted or may already have been transmitted but are still waiting for acknowledgment.

The frames outside the right side of the window cannot be sent until the window moves forward.

As acknowledgments arrive, the sender moves the window forward. This movement is called **sliding**.

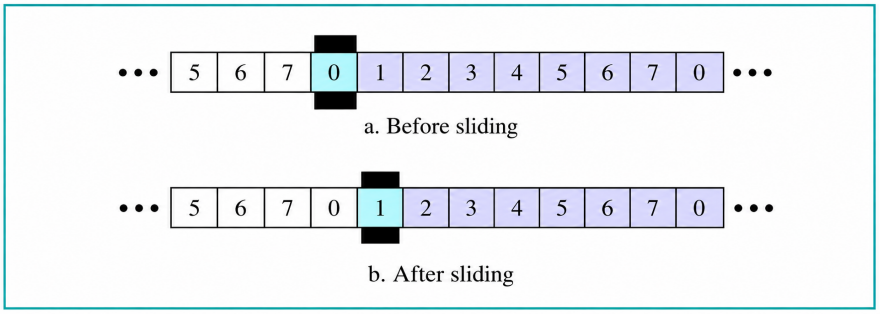
Figure 11.6 *Sender sliding window*



Receiver Sliding Window

The size of the window at the receiver site in Go-Back-N ARQ is always $W_R = 1$. The receiver is

Figure 11.7 *Receiver sliding window*

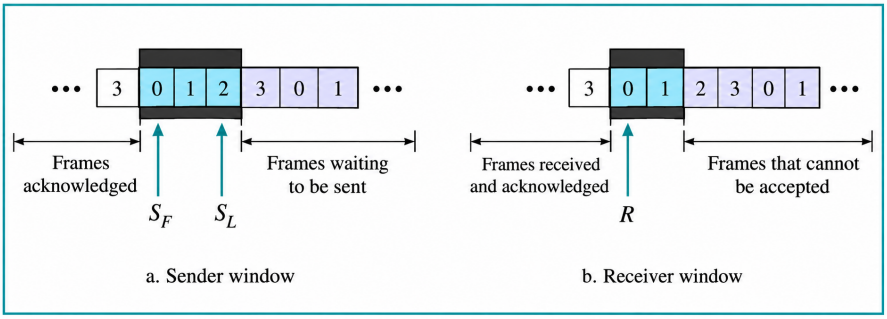


always looking for a specific frame to arrive in a specific order. If a frame arrives out of order, the receiver discards the frame and waits for the expected frame. For example, suppose the receiver is expecting Frame 2. If Frame 2 arrives it is accepted. If Frame 3 arrives before Frame 2 then Frame 3 discarded. The receiver continues to wait for Frame 2. After receiving the expected frame, the receiver's window moves to the next sequence number.

Control Variables: The sender uses three control variables: S_F, S, and S_L. The three variables are used to keep track of the sender window and the transmitted frames. S_F holds the sequence number of the first frame in the sender window, S holds the sequence number of the recently sent frame, and S_L holds the sequence number of the last frame in the sender window.The size of the sender window is $W = S_L - S_F + 1$. Because sequence numbers are cyclic, the calculation is interpreted modulo the available sequence-number space.

The receiver uses only one control variable: R. The variable R holds the sequence number of the

Figure 11.8 *Control variables*



frame the receiver expects to receive. If the sequence number of the received frame is equal to R, the

frame is accepted. If the received sequence number is not equal to R, the frame is rejected.

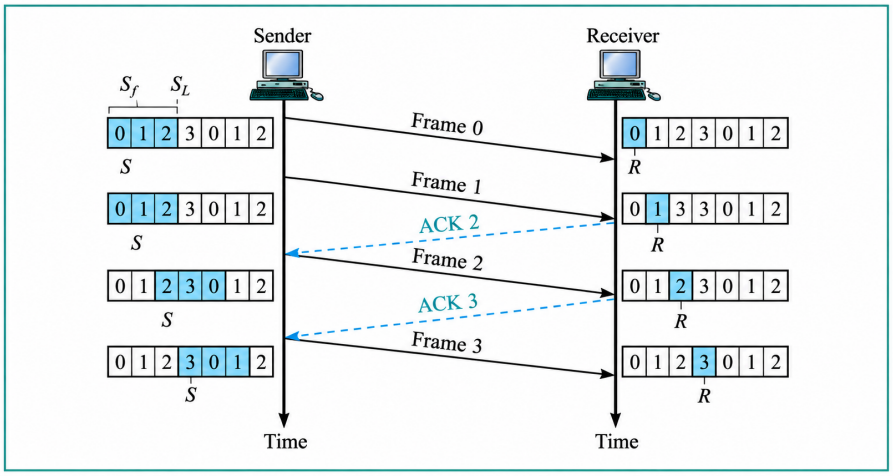
Acknowledgment: The receiver sends positive acknowledgments if a frame has arrived safe and sound and in order. If a frame is damaged or received out of order, the receiver is silent and will discard all subsequent frames until it receives the one it is expecting. The silence of the receiver causes the timer of the unacknowledged frame to expire. This, in turn, causes the sender to go back and resend all frames, beginning with the one with the expired timer. The receiver does not have to acknowledge each frame it receives. It can send one cumulative acknowledgment for several frames.

Resending Frame: When a frame is damaged, the sender goes back and sends a set of frames starting from the damaged one up to the last one sent. For example, suppose the sender has already sent frame 6, but the timer for frame 3 expires. This means that frame 3 has not been acknowledged, so the sender goes back and sends frames 3, 4, 5, 6 again. That is why the protocol is called Go-Back-N ARQ.

3.2 Operations

Normal Operation: Figure 11.9 shows normal operation of this mechanism.

Figure 11.9 *Go-Back-N ARQ, normal operation*



The sender keeps track of the outstanding frames and updates the variables and windows as the acknowledgments arrive.

Damaged or Lost Frame: Now let us see what happens if a frame is lost. Figure 11.10 shows that frame 2 is lost.

Note that when the receiver receives frame 3, it is discarded because the receiver is expecting frame 2, not frame 3 (according to its window). After the timer for frame 2 expires at the sender site, the sender sends frames 2 and 3 (it goes back to 2).

Delayed Acknowledgment: A delayed acknowledgment also triggers the resending of frames. Again, we leave the figure and details as an exercise.

Sender Window Size: We can now show why the size of the sender window must be less than 2^m. As an example, we choose $m = 2$, which means the size of the window can be $2^m - 1$, or 3. Figure 11.11 compares a window size of 3 and 4.

If the size of the window is 3 (less than 2^2) and all three acknowledgments are lost, the frame 0 timer expires and all three frames are sent. However, the window of the receiver is now expecting frame 3, not frame 0, so the duplicate frame is correctly discarded.

Figure 11.10 *Go-Back-N ARQ, lost frame*

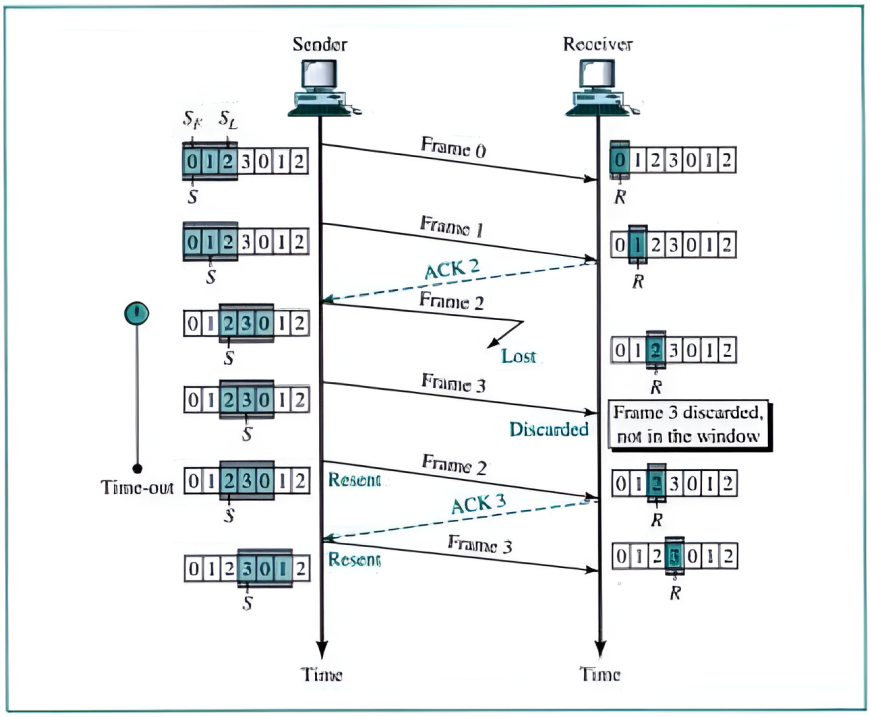
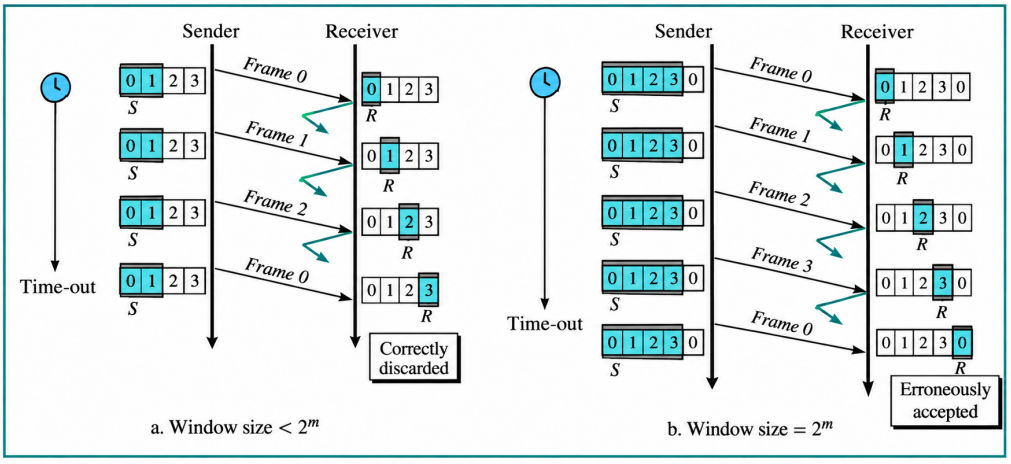


Figure 11.11 *Go-Back-N ARQ: sender window size*



On the other hand, if the size of the window is 4 (equal to 2^2) and all acknowledgments are lost, the sender will send the duplicate of frame 0. However, this time the window of the receiver expects to receive frame 0, so it accepts frame 0, not as a duplicate, but as the first frame in the next cycle. This is an error.

Therefore, the size of the sender window in Go-Back-N ARQ must satisfy $W < 2^m$. Hence, the maximum sender window size is $W_{\max} = 2^m - 1$

4 Selective Repeat ARQ

Go-Back-N ARQ simplifies the process at the receiver site. The receiver keeps track of only one variable, and there is no need to buffer out-of-order frames; they are simply discarded.

However, this protocol is very inefficient for a noisy link. In a noisy link, a frame has a high probability of damage, which means the resending of multiple frames. This resending uses up the bandwidth and slows down the transmission.

For noisy links, there is another mechanism that does not resend N frames when just one frame

is damaged; only the damaged frame is resent. This mechanism is called **Selective Repeat ARQ**. It is more efficient for noisy links, but the processing at the receiver is more complex.

4.1 Sender and Receiver Windows

The configuration of the sender and its control variables for Selective Repeat ARQ are the same as for Go-Back-N ARQ.

However, for reasons to be discussed later, the size of the window should be at most one-half of the value 2^m. Thus, $W \leq \frac{2^m}{2}$ or $W \leq 2^{m-1}$. The receiver window must also be this size.

This window, however, specifies the range of the accepted received frames. In other words, in Go-Back-N, the receiver is looking for one specific sequence number; in Selective Repeat, the receiver is looking for a range of sequence numbers.

The receiver has two control variables, R_F and R_L, to define the boundaries of the window.

Figure 11.12 *Selective Repeat ARQ, sender and receiver windows*

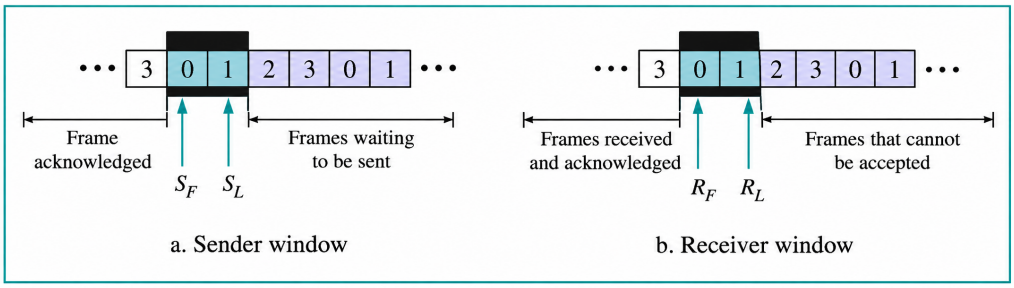


Figure 11.12 shows the sender and receiver windows.

Negative Acknowledgment: Selective Repeat ARQ also defines a **negative acknowledgment (NAK)** that reports the sequence number of a damaged frame before the timer expires.

4.2 Operation

Let us show the operation of the mechanism with an example of a lost frame, as shown in Figure 11.13.

Frames 0 and 1 are accepted when received because they are in the range specified by the receiver window. When frame 3 is received, it is also accepted for the same reason. However, the receiver sends a NAK 2 to show that frame 2 has not been received. When the sender receives the NAK 2, it resends only frame 2, which is then accepted because it is in the range of the window.

Lost and Delayed ACKs and NAKs

We leave lost and delayed ACKs and NAKs as exercises. Note that the sender also sets a timer for each frame sent.

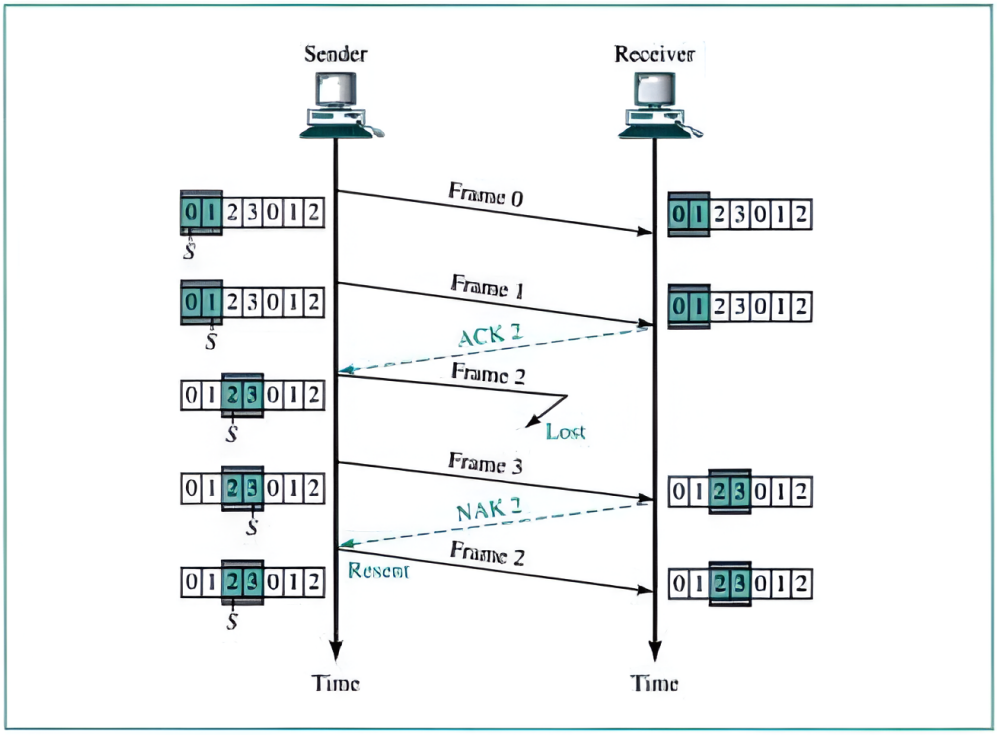
Sender Window Size:

We can now show why the size of the sender and receiver windows must be at most one-half of 2^m. For example, we choose $m = 2$, which means the size of the window can be one-half of 2^2, or 2. Figure 11.14 compares a window size of 2 with a window size of 3.

If the size of the window is 2 and all acknowledgments are lost, the timer for frame 0 expires and frame 0 is resent. However, the window of the receiver is now expecting frame 2, not frame 0, so this duplicate frame is correctly discarded.

When the size of the window is 3 and all acknowledgments are lost, the sender sends a duplicate of frame 0. However, this time, the window of the receiver expects to receive frame 0 (0 is part of the

Figure 11.13 *Selective Repeat ARQ, lost frame*



window), so it accepts frame 0, not as a duplicate, but as the first frame in the next cycle. This is an error.

Therefore, in Selective Repeat ARQ, the size of the sender and receiver window must be at most one-half of 2^m. That is $W \leq \frac{2^m}{2}$ or $W \leq 2^{m-1}$.

Figure 11.14 *Selective Repeat ARQ, sender window size*

